

APPLIED ARTIFICIAL NEURAL NETWORK



Chapter 1

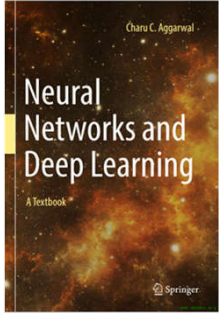
Introduction to neural network

1.1 Biological neural network

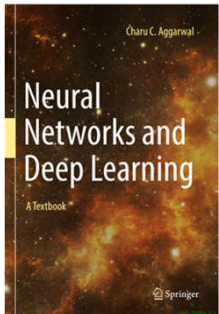
1.2 Artificial neural network

1.3 Applications of artificial neural networks

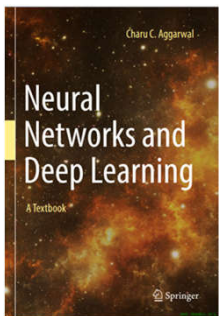
REFERENCES



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Michael Nielsen, *Neural Networks and Deep Learning*, Google Book, 2019



Ahmed Gad, Fatima Jarmouni, *Introduction to Deep Learning and Neural Networks with Python*, Elsevier, 2020



1.1 Biological neural network

1.1 Biological neural network

- The human nervous system is a complex network of cells, which are referred to as neurons.
- The human brain is a complex, nonlinear, and parallel information processing system, capable of learning, memory and generalization.
- The human brain is made up of about 10^{11} neurons, which linked together into a network



1.1 Biological neural network

- A typical neuron consists of a cell body (soma), dendrites, and a single axon.

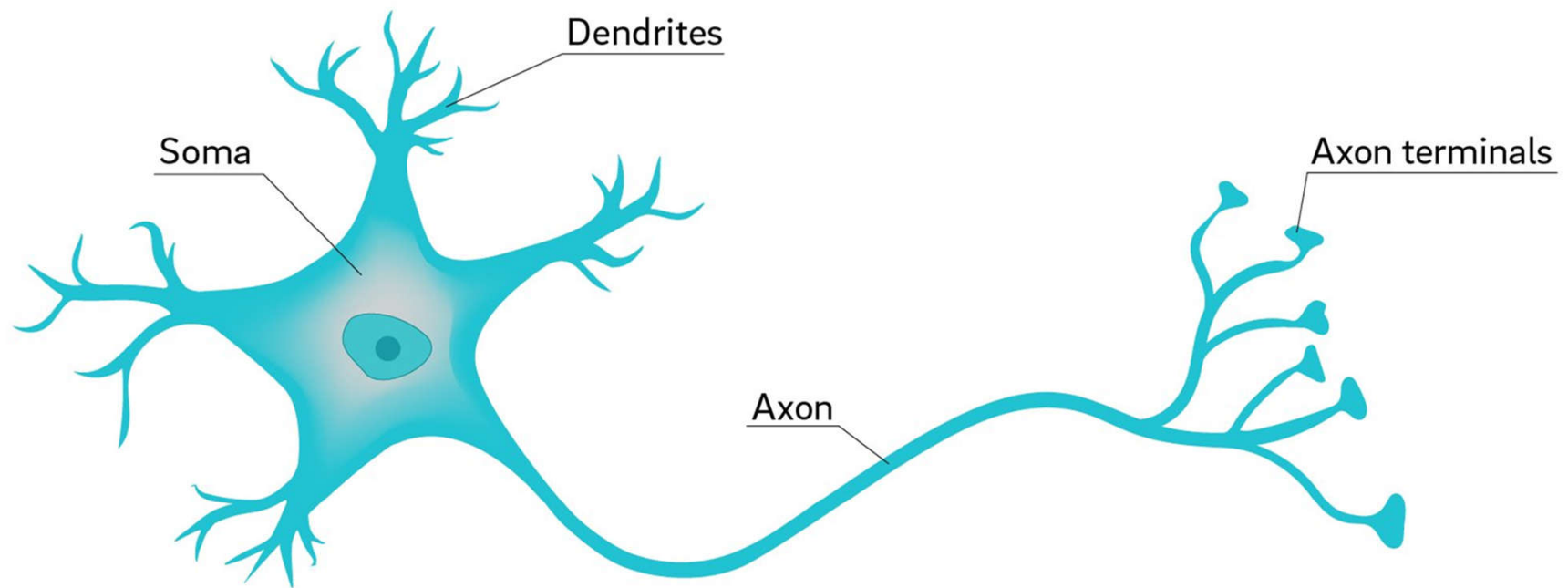


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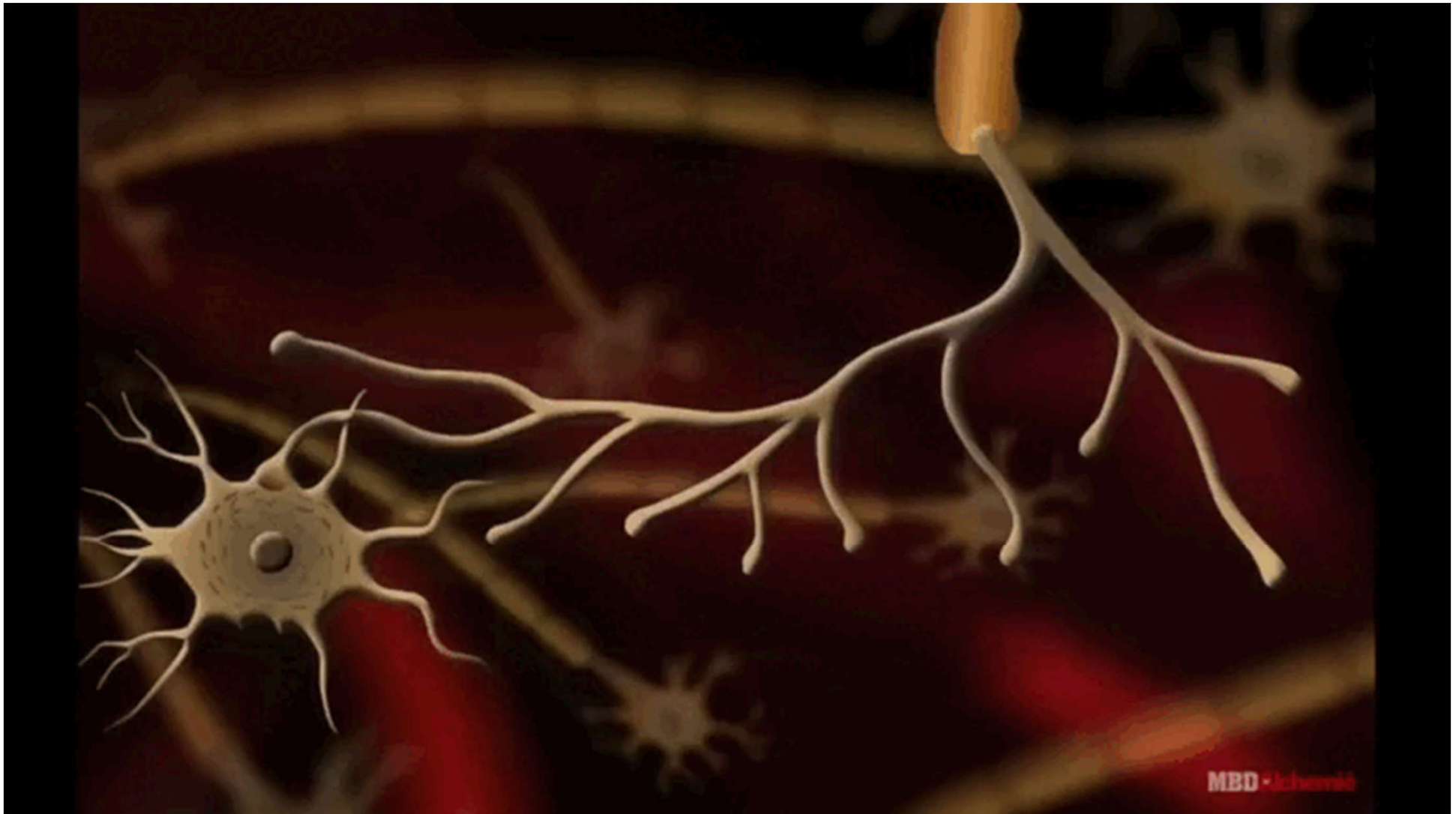
1.1 Biological neural network

How BNN work

- Each neuron receives electrical signals (impulses) from neighboring neurons via fibers called *dendrites*.
- *Cell body (soma)* has the function of synthesizing incoming signals.
- When the total sum of incoming signals exceeds some *threshold*, the neuron “fires” output signal via long fibers called *axons*.
- The junction between two neurons is called a *synapse*.

1.1 Biological neural network

How BNN work





1.2 Artificial neural network

1.2 Artificial neural network

What is ANN

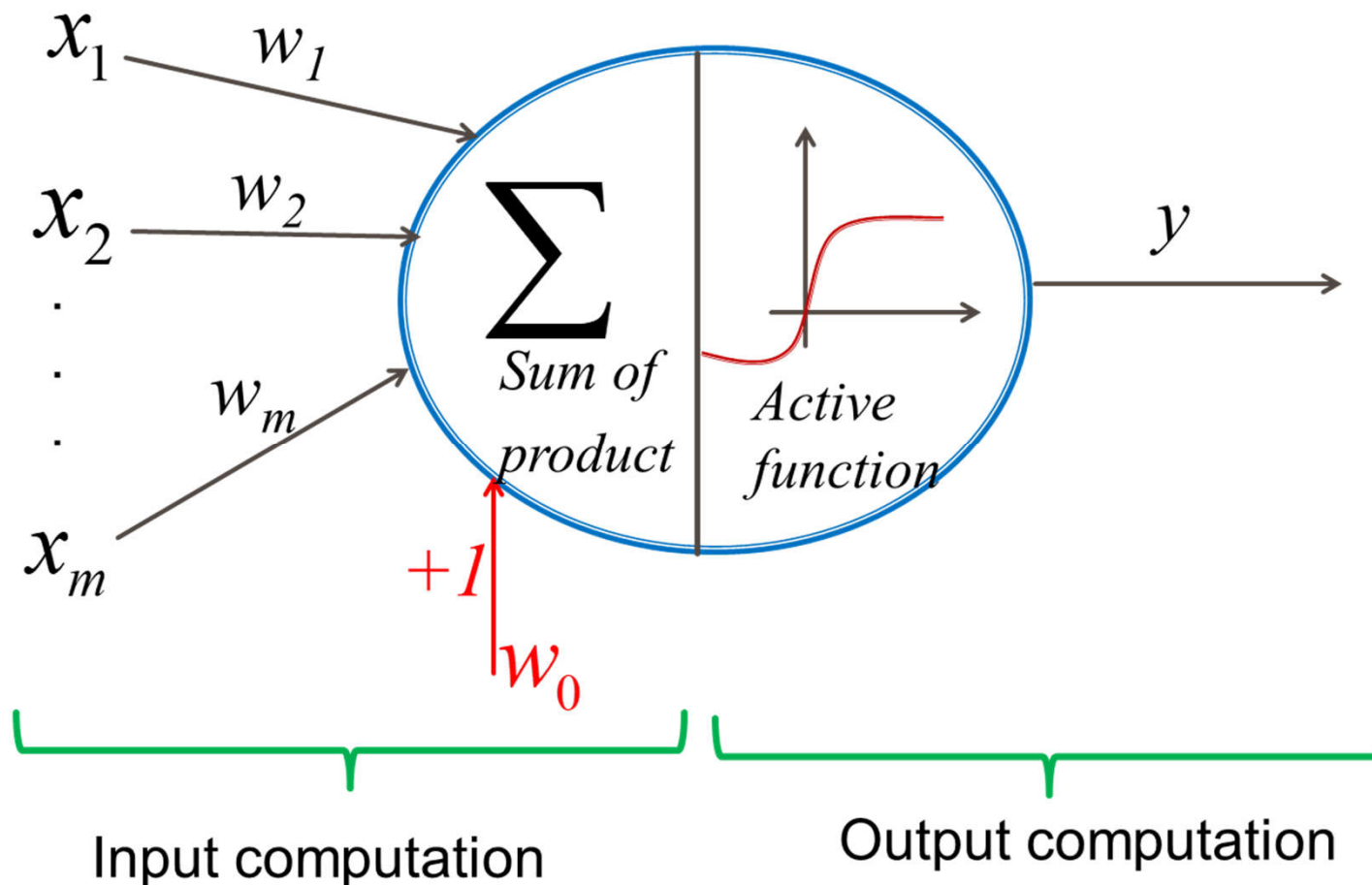
- Artificial neural network (ANN) simulate the human–brain processes by mathematic.

Biological Neural Network	Artificial Neural Network
Dendrites	Inputs
Cell nucleus	Nodes
Synapse	Weights
Axon	Output

1.2 Artificial neural network

1.2.1 Single layer neural network, perceptron

➤ Neural network computation consists of two processes:



1.2 Artificial neural network

Input computation

➤ The equation is expressed in summation form as.

$$f = net = \sum_{i=1}^n w_i x_i + w_0$$

or in vector form as

$$f = net = \mathbf{W}^T \mathbf{X} + w_0$$

Where $\mathbf{W}$, $\mathbf{X}$ are m -dimension column vector; w_0 – bias

$$\mathbf{W} = \begin{bmatrix} w_1 \\ w_2 \\ . \\ . \\ w_m \end{bmatrix} \quad \text{- vector of weights} \quad \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ . \\ . \\ x_m \end{bmatrix} \quad \text{- input vector with } m\text{-features}$$

1.2 Artificial neural network

Input computation

- Expanding the weight vector and input vector

$$\mathbf{W} = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \cdot \\ \cdot \\ w_m \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_m \end{bmatrix}$$

The computation can be rewritten as follows

$$f = net = \sum_{i=0}^m w_i x_i = \mathbf{W}^T \mathbf{X}$$

Where $\mathbf{W}$, $\mathbf{X}$ are $m+1$ - dimension column vector.

1.2 Artificial neural network

Output computation

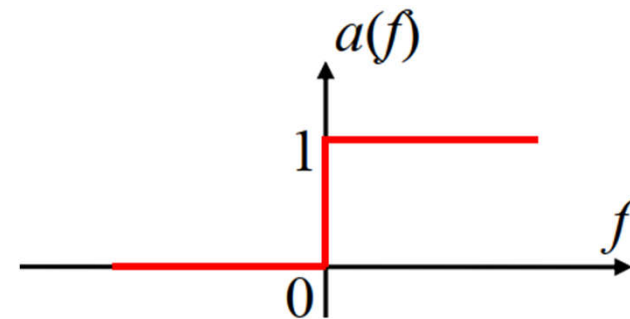
- The output of the neural, denoted by y , is obtained by passing f (or net) through the active function a .

$$y = a(f)$$

- Some active functions:

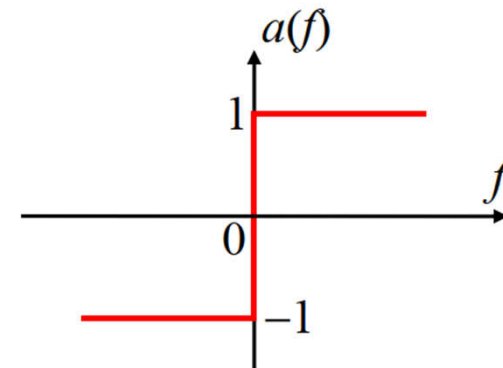
1. Step function:

$$y = a(f) = \begin{cases} 1 & \text{if } f \geq 0 \\ 0 & \text{if } f < 0 \end{cases}$$



2. Sign function:

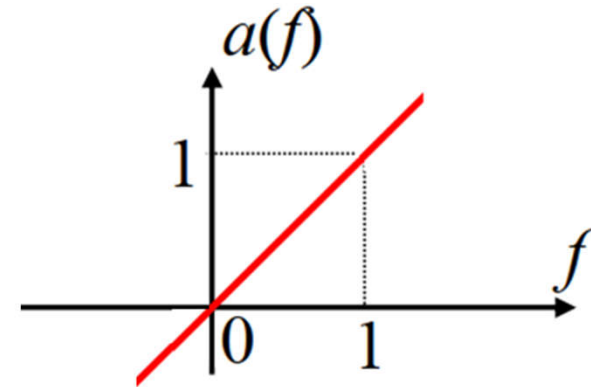
$$y = a(f) = \begin{cases} +1 & \text{if } f \geq 0 \\ -1 & \text{if } f < 0 \end{cases}$$



1.2 Artificial neural network

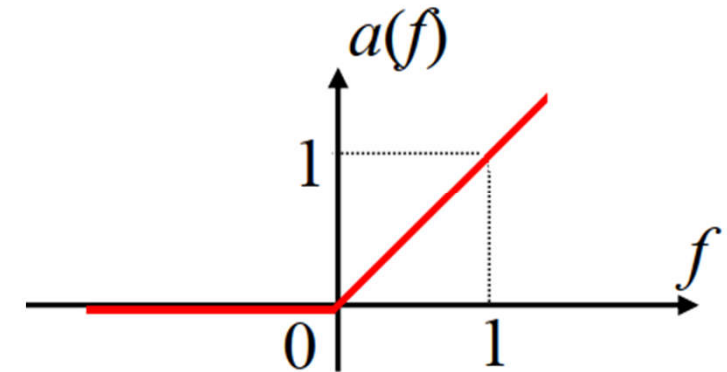
3. *Linear function:*

$$y = a(f) = f$$



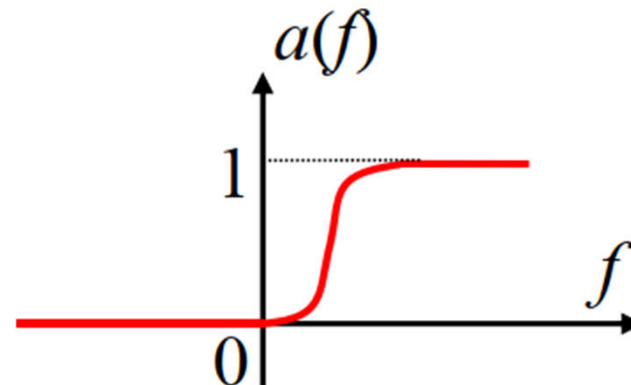
4. *Rectifier Linear Unit (ReLU):*

$$y = a(f) = \begin{cases} f & \text{if } f \geq 0 \\ 0 & \text{if } f < 0 \end{cases}$$



5. *Sigmoid Function*

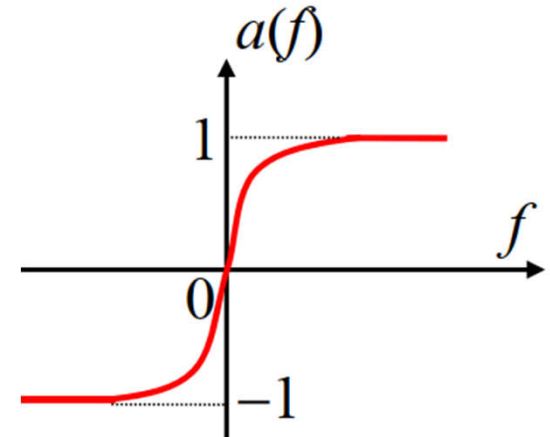
$$y = a(f) = \frac{1}{1 + e^{-f}}$$



1.2 Artificial neural network

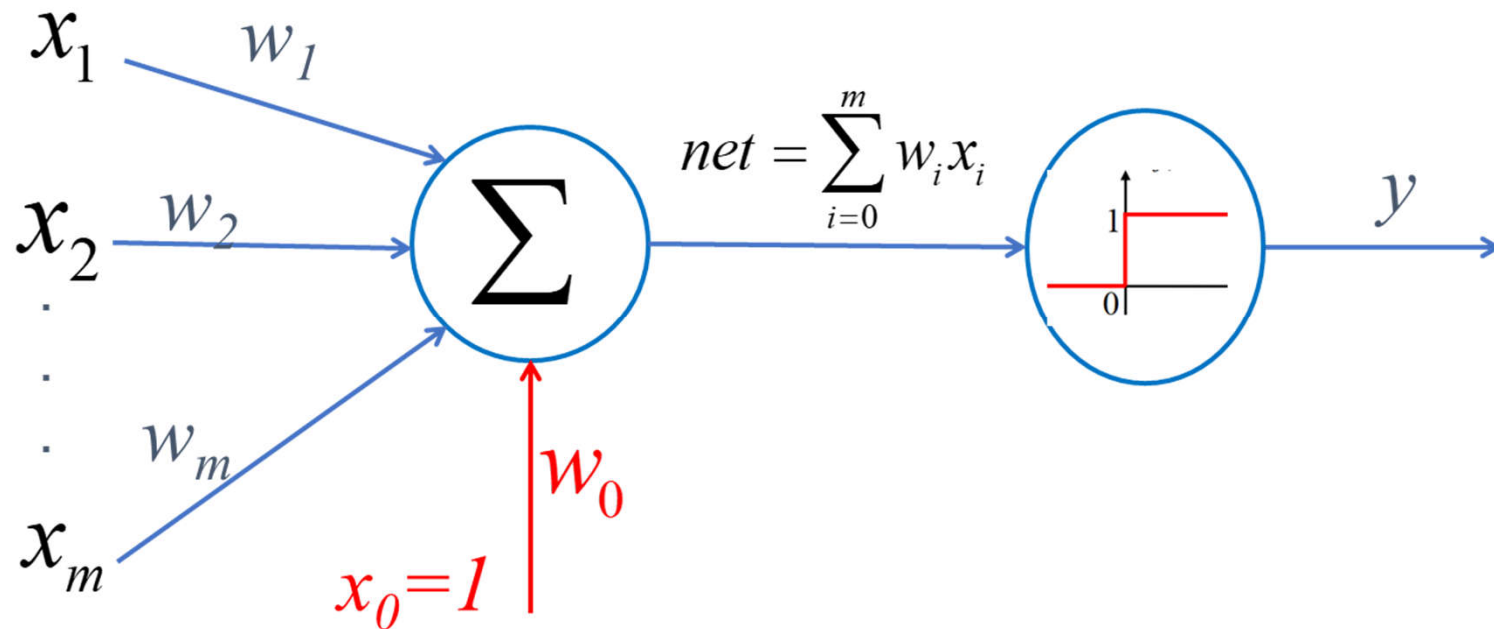
6. Hyperbolic tangent Function

$$y = a(f) = \tanh(f)$$



- The simplest single layer neural network is referred to as *the perceptron*.
- The perceptron contains a single input layer and an output node.
- Output of perceptron is a binary value.

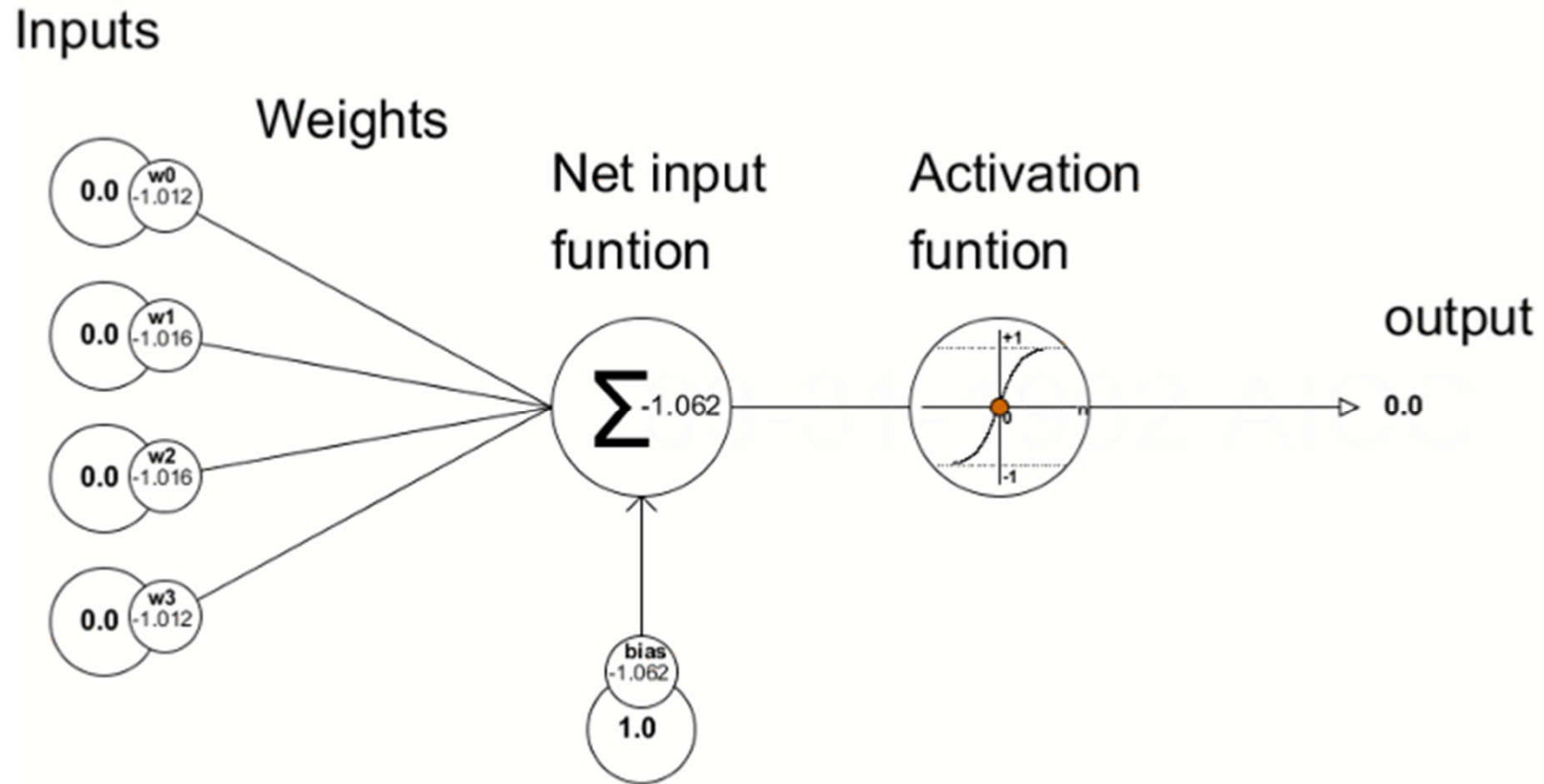
1.2 Artificial neural network



The perceptron model

- In perceptron, the active function is **sign function** or **step function**.

1.2 Artificial neural network

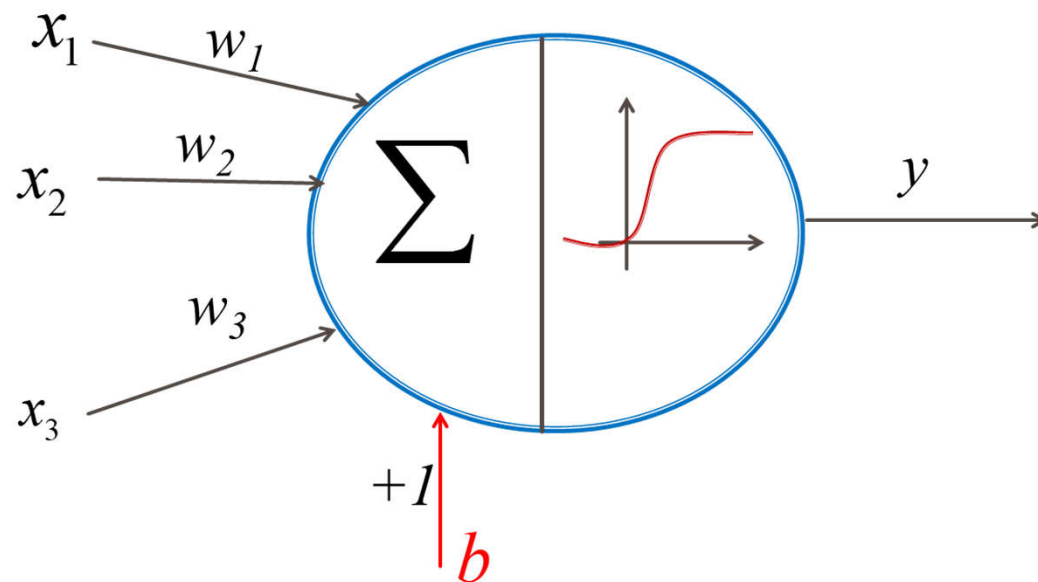


Single neuron in action. Source [here](#)

1.2 Artificial neural network

Example:

- Computation the output of the neural y , with the sigmoid active function.



$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.5 \\ 1 \end{bmatrix} \quad \mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 0.3 \\ -0.8 \\ 0.4 \end{bmatrix} \quad b = 0.6$$

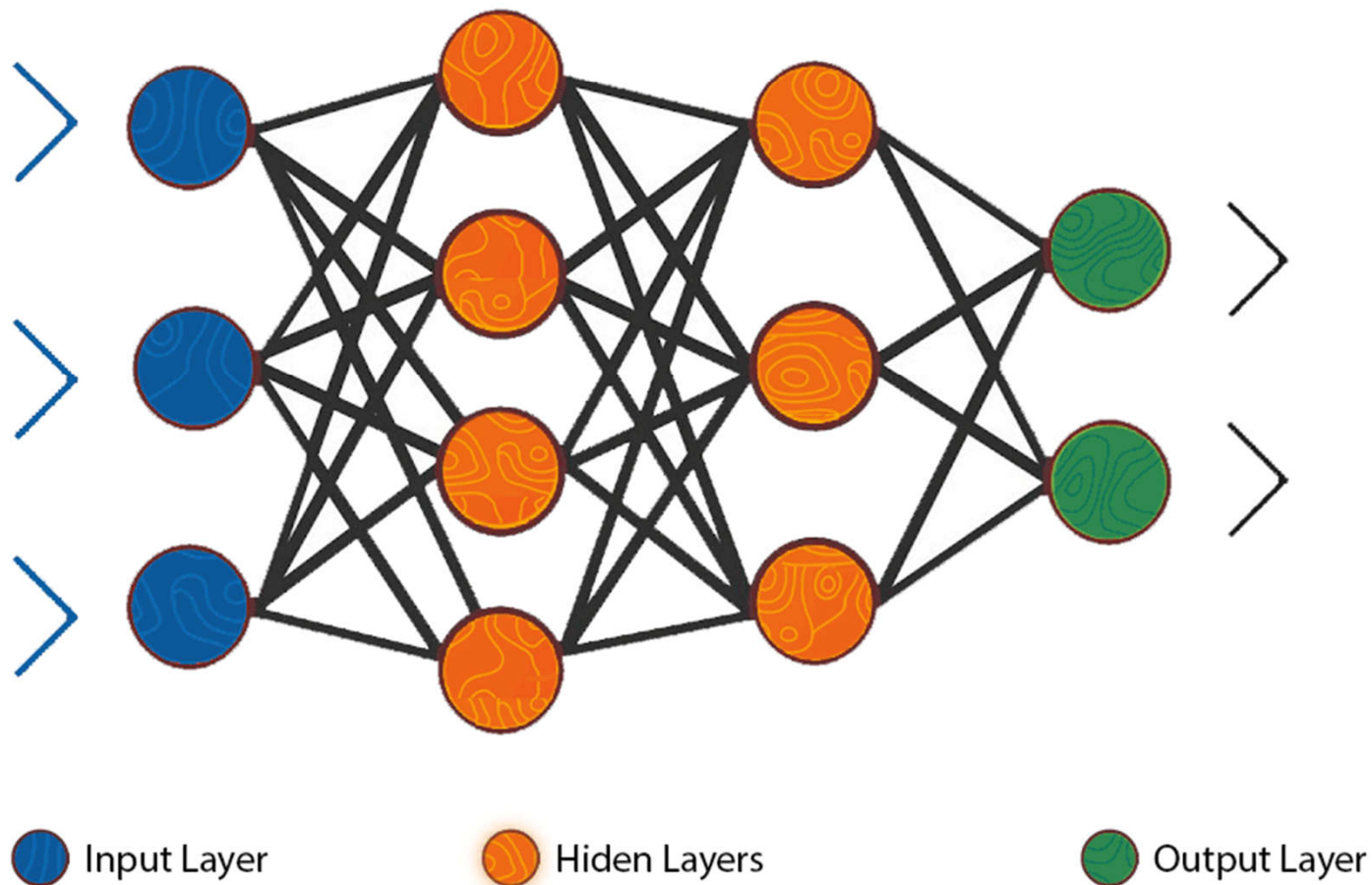
1.2 Artificial neural network

1.2.2 Multi-layer neural network

- The neurons are linked together to form a neural network.
- The neural network consists of 3 layers: input layer, hidden layer and output layer.
- The neural network with a single hidden layer is called **shallow neural network**, and refer to network with two or more hidden layers as a **deep neural network**.

1.2 Artificial neural network

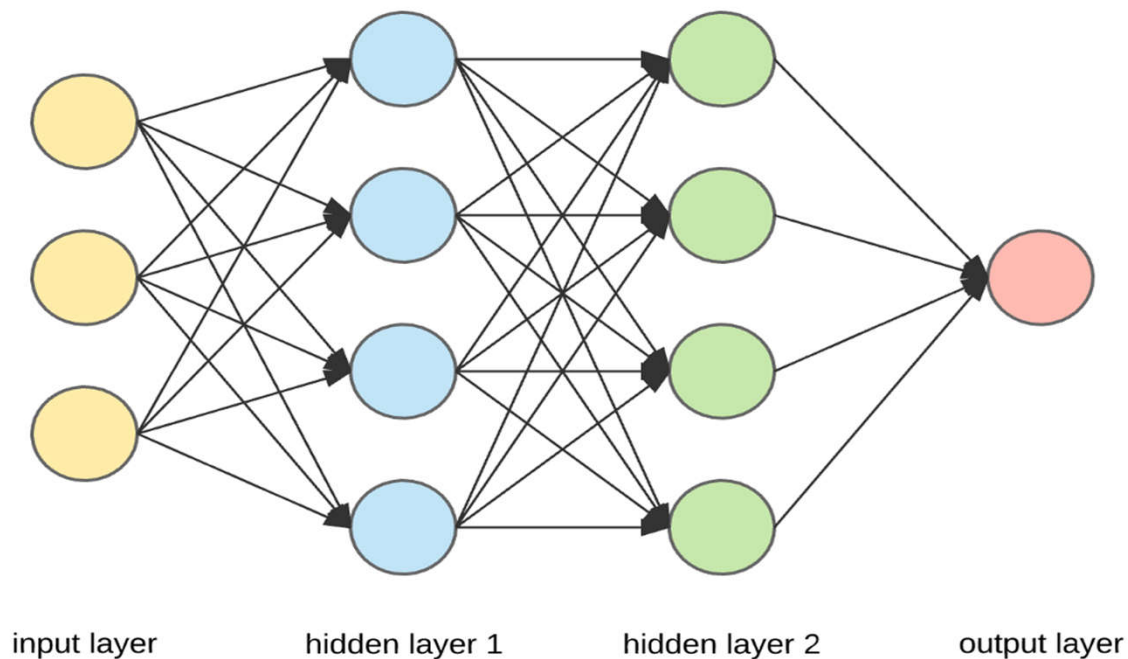
1.2.2 Multi-layer neural network



1.2 Artificial neural network

1.2.2 Multi-layer neural network

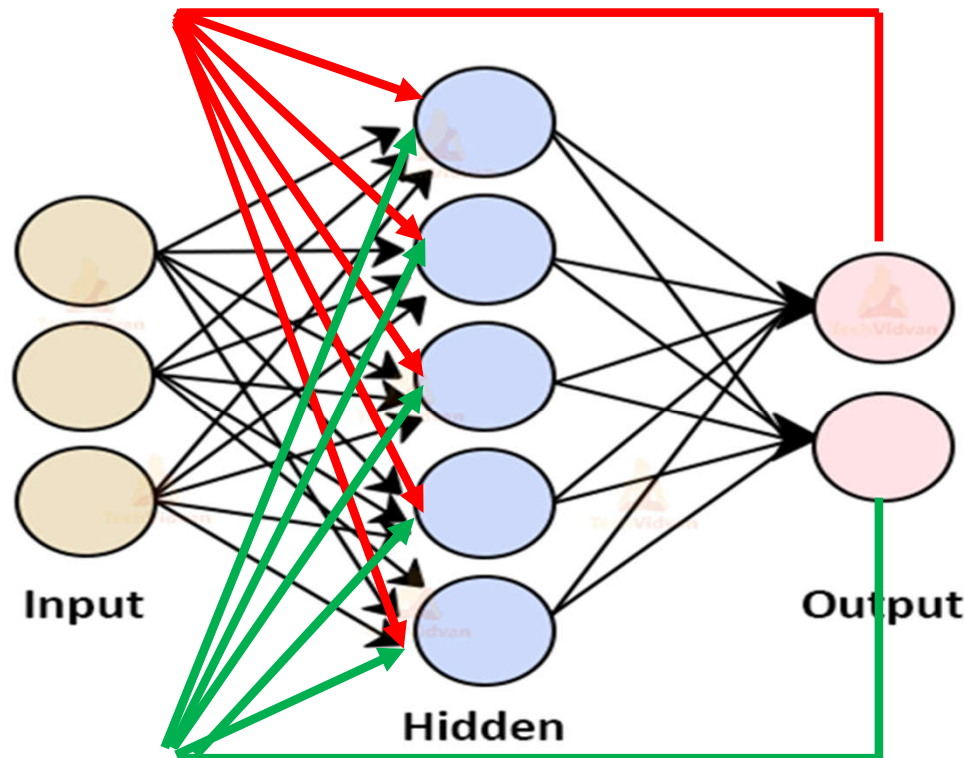
- **Feed-Forward Networks** - A multi-layered neural network where the information moves from left to right, or in other words, in a forward direction. The input values pass through a series of hidden layers on their way to the output layer.



1.2 Artificial neural network

1.2.2 Multi-layer neural network

- **Recurrent neural networks**, also known as RNNs, are a class of neural networks that allow previous outputs to be used as inputs while having hidden states.



1.2 Artificial neural network

NEURAL NETWORK COMPUTATION

- The outputs of the layer 1 are the components of input vector $\mathbf{x}$:

$$z_j = x_j \quad j = 1, 2, \dots, m$$

The matrix form: $\mathbf{Z}(1) = \mathbf{X}$

- The matrix weights in layer l :

$$\mathbf{W}(\ell) = \begin{bmatrix} w_{11}(\ell) & w_{12}(\ell) & \cdots & w_{1n_{\ell-1}}(\ell) \\ w_{21}(\ell) & w_{22}(\ell) & \cdots & w_{2n_{\ell-1}}(\ell) \\ \vdots & \vdots & \cdots & \\ w_{n_{\ell}1}(\ell) & w_{n_{\ell}2}(\ell) & \cdots & w_{n_{\ell}n_{\ell-1}}(\ell) \end{bmatrix}$$

1.2 Artificial neural network

NEURAL NETWORK COMPUTATION

- Obtain all the sum-of-products computations, $net_i(l)$, for layer l simultaneously:

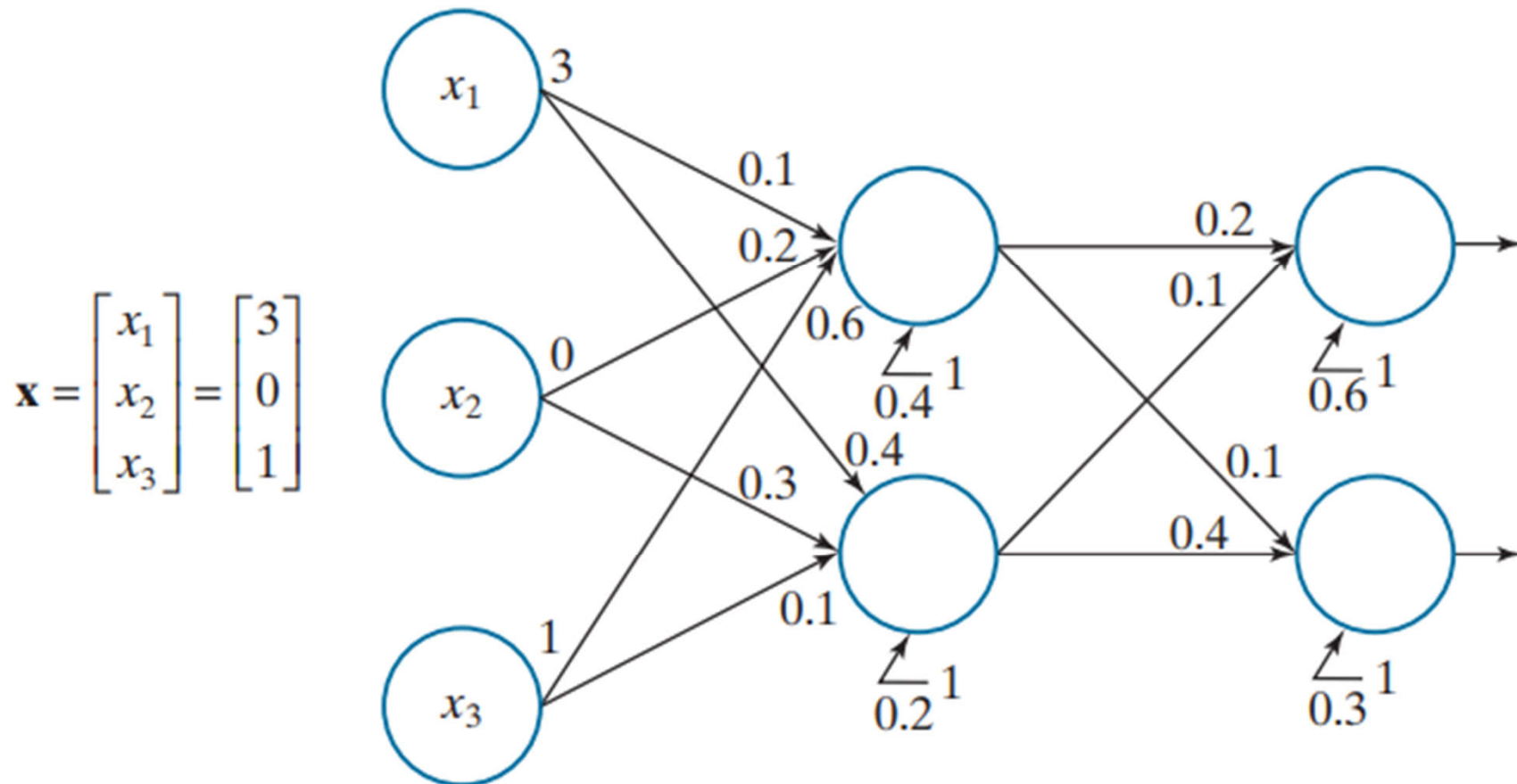
$$\mathbf{net}(l) = \mathbf{W}(l) * \mathbf{Z}(l-1) + \mathbf{b}(l) \quad l = 2, 3, \dots, L$$

- Because the activation function is applied to each net input independently of the others, the outputs of the network at any layer can be expressed in vector form as:

$$\mathbf{Z}(l) = a[\mathbf{net}(l)] = \begin{bmatrix} a(net_1(l)) \\ a(net_2(l)) \\ \dots \\ a(net_{nl}(l)) \end{bmatrix}$$

1.2 Artificial neural network

Example





1.3 Applications of Artificial neural network

1.3 Applications of Artificial neural network

Artificial neural networks (ANNs) have a wide range of applications across various fields due to their ability to learn from data and perform complex tasks. Some notable applications of artificial neural networks include:

1. Perceptron

- Classification.
- Encode Database (Multilayer Perceptron).
- Monitor Access Data (Multilayer Perceptron).

1.3 Applications of Artificial neural network

2. Feed Forward (FF)

- Data Compression.
- Pattern Recognition.
- Computer Vision.
- Sonar Target Recognition.
- Speech Recognition.
- Handwritten Characters Recognition.

1.3 Applications of Artificial neural network

3. Radial Basis Network (RBN)

- Function Approximation.
- Timeseries Prediction.
- Classification.
- System Control.

4. Long / Short Term Memory (LSTM)

- Speech Recognition.
- Writing Recognition.

1.3 Applications of Artificial neural network

5. Recurrent Neural Network (RNN)

- Machine Translation.
- Robot Control.
- Time Series Prediction.
- Speech Recognition.
- Speech Synthesis.
- Time Series Anomaly Detection.
- Rhythm Learning.
- Music Composition.

1.3 Applications of Artificial neural network

6. Markov Chain (MC)

- Speech Recognition.
- Information And Communication System.
- Queuing Theory.
- Statistics.

1.3 Applications of Artificial neural network

7. Convolutional Neural Networks

- Identify Faces, Street Signs, Tumors.
- Image Recognition.
- Video Analysis.
- NLP.
- Anomaly Detection.
- Drug Discovery.
- Checkers Game.
- Time Series Forecasting.

1.3 Applications of Artificial neural network

8. Support Vector Machines (SVM)

- Face Detection.
- Text Categorization.
- Classification.
- Bioinformatics.
- Handwriting recognition.